

Netec1400 XAUUSD Grid / Recovery EA

Overview

This Expert Advisor (EA) implements a grid/recovery trading system specifically designed for XAUUSD (Gold) on MetaTrader 4. The EA prioritizes **survivability over profit**, with comprehensive risk controls and drawdown management.

Priority Order: Survivability > Control > Stability > Profit

Files Delivered

- **Netec1400_XAUUSD_Grid.mq4** - Full MQ4 source code
- **Netec1400_XAUUSD_Grid.ex4** - Compiled executable (compile via MetaEditor)
- **README.md** - This documentation

Core Features

1. Entry Logic (Phase-1: Manual and Semi-Auto Only)

Phase-1 entry is **controllable only**; no autonomous indicator-based entry. The EA supports:

- **Mode 0 (Manual – Adopt):** The EA does **not** open the first trade. You place the first order(s) in MT4 (same symbol, EA MagicNumber). The EA detects them and **adopts** them as the basket start, then manages grid/recovery. Single direction only; mixed BUY/SELL orders are rejected.
- **Mode 1 (Semi-Auto):** The EA can show a **suggested** direction (e.g. from MA, display only). You set **OpenBasketDirection** to BUY or SELL to confirm. When filters pass, the EA opens the first trade in that direction. No automatic opening from indicators.

Indicator-based entry (MA/RSI/Time) is **out of scope for Phase-1** and deferred to a later phase; that code remains in source but is not used for opening trades in Phase-1.

2. Grid Logic

- **Single Direction Basket:** All orders in a basket are the same direction (BUY or SELL)
- **Fixed Grid Distance:** Adds new grid level when price moves against basket by **GridDistancePoints**
- **Dynamic Grid Expansion:** When basket drawdown reaches **IncreaseGridDistanceAtDDPercent**, grid distance increases by **DynamicGridMultiplier**
- **Maximum Levels:** Enforces **MaxGridLevels** to prevent unlimited grid expansion
- **Controlled Lot Sizing:** Uses fixed lots by default; optional controlled progression via **MaxLotMultiplier** (no aggressive martingale)

3. Risk & Drawdown Control (ProtectionMode / EmergencyExit)

The EA implements multiple layers of risk protection:

ProtectionMode (when limits are approached)

- **Pause New Trades:** When basket DD $\geq$ `PauseNewTradesAtDDPercent`, stops opening new grid levels
- **Increase Grid Distance:** When basket DD $\geq$ `IncreaseGridDistanceAtDDPercent`, expands grid spacing
- No blind liquidation; behavior is controlled and configurable

EmergencyExit

- **Emergency Close:** When basket DD $\geq$ `EmergencyCloseAtDDPercent`, closes entire basket in a **controlled** way (slippage limit, retries)
- **EmergencyExit** is configurable and designed to minimize damage, not force liquidation

Account-Level Controls

- **Equity Drawdown Limit:** Blocks new baskets when equity DD $\geq$ `MaxEquityDrawdownPercent`
- **Margin Protection:** Requires free margin $\geq$ `MinFreeMarginPercent` before any order
- **Emergency Margin Close:** Triggers `EmergencyExit` if free margin drops below 50% of minimum

4. Market Protection Filters

- **Spread Filter:** Blocks trading when spread $>$ `MaxSpreadPoints`
- **Volatility Filter:** Uses ATR to detect abnormal volatility
 - Blocks new baskets when $ATR > (20\text{-period average} \times ATR_MultiplierThreshold)$
 - Pauses grid expansion during high volatility
- **Trading Hours:** Optional session filter via `StartHour` and `EndHour`

5. Exit Logic

- **Basket Take Profit:** Closes entire basket when total profit $\geq$ `BasketTakeProfitMoney`
- **Trailing Basket TP** (optional): Locks in profit by trailing the basket's highest profit level
 - Activates when profit reaches `TrailingStart`
 - Closes basket if profit falls back by `TrailingStep`
- **EmergencyExit:** Controlled, configurable basket closure with slippage limits during extreme conditions; minimizes damage, no blind liquidation

6. Recovery & State Management

- **Auto-Recovery:** Restores basket state on EA restart if orders exist
- **State Consistency:** Automatically detects and resets state if orders are manually closed
- **Clean Restart:** After basket closure (normal or emergency), EA cleanly returns to "no basket" state

Key Input Parameters

Essential Settings

Parameter	Default	Description
<code>MagicNumber</code>	100140	Unique identifier for EA orders
<code>EntryMode</code>	0	Phase-1: 0 = Manual (adopt user order), 1 = Semi-Auto (user confirms via input)

Parameter	Default	Description
OpenBasketDirection	0	Semi-Auto only: 0 = None, 1 = BUY, 2 = SELL (confirm to open basket)
LotSize	0.01	Initial lot size
MaxGridLevels	10	Maximum grid depth
GridDistancePoints	500	Distance between grid levels (points)
BasketTakeProfitMoney	10.0	Basket TP in account currency

Risk Parameters

Parameter	Default	Description
MaxBasketDrawdownMoney	100.0	Maximum basket DD in account currency
MaxEquityDrawdownPercent	20.0	Maximum account equity DD (%)
PauseNewTradesAtDDPercent	50.0	Pause grid at this basket DD (%)
IncreaseGridDistanceAtDDPercent	30.0	Expand grid at this basket DD (%)
EmergencyCloseAtDDPercent	80.0	Emergency close at this basket DD (%)
MinFreeMarginPercent	30.0	Minimum free margin required (%)

Filter Parameters

Parameter	Default	Description
MaxSpreadPoints	50	Maximum allowed spread (points)
UseVolatilityFilter	true	Enable ATR volatility filter
ATR_Period	14	ATR calculation period
ATR_MultiplierThreshold	2.0	Volatility threshold multiplier

Implementation Details

Basket Identification

- All orders with the same **MagicNumber** and same order type (OP_BUY or OP_SELL) belong to one basket
- Only one basket can be active at a time
- No mixed-direction baskets (all BUY or all SELL)

Order of Safety Checks

Before opening any order, the EA checks in this order:

1. Symbol must be XAUUSD
2. Spread $\leq$ **MaxSpreadPoints**

3. Volatility acceptable (if filter enabled)
4. Within trading hours (if filter enabled)
5. Equity drawdown < limit
6. Free margin $\geq$ minimum
7. Basket drawdown < pause threshold (for grid levels)
8. Maximum grid levels not reached (for grid levels)

Emergency Close Behavior

Emergency close is **controlled**, not a blind liquidation:

- Closes orders one by one with retry logic (up to 3 attempts)
- Respects **MaxBasketSlippagePoints** slippage limit
- Logs all actions for review
- Resets basket state after closure

Edge Case Handling

- **Requotes**: Retries OrderSend with delay
- **Manual Close**: Detects and resets state when orders are manually closed
- **Broker Digits**: Uses **MarketInfo()** and **Point** for proper point/pip calculations
- **Weekend Gaps**: No new orders outside market hours; exit logic can still run

Installation & Usage

Installation

1. Copy **Netec1400_XAUUSD_Grid.mq4** to [MT4 Data Folder]/MQL4/Experts/
2. Open MetaEditor (F4 in MT4)
3. Open the MQ4 file and click "Compile" (F7)
4. Verify **Netec1400_XAUUSD_Grid.ex4** is created in the same folder

Attaching to Chart

1. Open an XAUUSD chart in MT4
2. Drag the EA from Navigator onto the chart
3. Configure input parameters
4. Enable "Allow live trading" and "Allow DLL imports" (if needed)
5. Click OK

Testing Workflow

IMPORTANT: Follow this sequence before live trading:

1. **Strategy Tester** (MT4 built-in):
 - Test on XAUUSD M1 or H1 timeframe
 - Use quality data (99% modeling quality)
 - Test various market conditions (trending, ranging, volatile)
 - Verify all risk controls activate correctly

2. **Demo Account:**

- Run on demo for at least 1-2 weeks
- Monitor basket behavior, grid expansion, and exits
- Verify emergency close works as expected
- Check that DD limits are respected

3. **Live Account** (only after approval):

- Start with minimum lot sizes
- Monitor closely for first few baskets
- Gradually increase lot size if behavior is stable

Configuration Recommendations

Conservative Settings (Recommended for Start)

```
LotSize = 0.01
MaxGridLevels = 5
GridDistancePoints = 800
BasketTakeProfitMoney = 5.0
MaxBasketDrawdownMoney = 50.0
MaxEquityDrawdownPercent = 10.0
PauseNewTradesAtDDPercent = 40.0
EmergencyCloseAtDDPercent = 70.0
```

Moderate Settings

```
LotSize = 0.01
MaxGridLevels = 10
GridDistancePoints = 500
BasketTakeProfitMoney = 10.0
MaxBasketDrawdownMoney = 100.0
MaxEquityDrawdownPercent = 20.0
PauseNewTradesAtDDPercent = 50.0
EmergencyCloseAtDDPercent = 80.0
```

Aggressive Settings (Higher Risk)

```
LotSize = 0.02
MaxGridLevels = 15
GridDistancePoints = 300
BasketTakeProfitMoney = 20.0
MaxBasketDrawdownMoney = 200.0
MaxEquityDrawdownPercent = 30.0
PauseNewTradesAtDDPercent = 60.0
EmergencyCloseAtDDPercent = 90.0
```

Monitoring & Logs

On-Chart Display

The EA displays real-time information via `Comment()`:

- Magic number
- Current basket direction (BUY/SELL/NONE)
- Number of open orders vs maximum
- Current basket profit
- Basket drawdown percentage
- Equity drawdown percentage
- Current spread
- Free margin percentage

Log Messages

The EA logs all important events to the Experts log:

- Basket opened (direction, price, lot)
- Grid level added (level number, price, DD%)
- Basket closed (reason, profit)
- Emergency close triggered (reason)
- Blocked actions (spread, volatility, DD, margin)

Always review logs after testing and during live operation.

Troubleshooting

EA Not Opening Trades

Check in order:

1. Symbol is XAUUSD
2. Spread is acceptable (check `MaxSpreadPoints`)
3. Volatility filter not blocking (check ATR vs threshold)
4. Within trading hours (if `UseTradingHours = true`)
5. Equity DD below limit
6. Free margin above minimum
7. Entry signal present (check `EntryMode` and indicator values)

Grid Not Expanding

Check:

1. Basket DD < `PauseNewTradesAtDDPercent`
2. Spread acceptable
3. Volatility acceptable (if filter enabled)
4. Free margin sufficient

5. Current orders < `MaxGridLevels`
6. Price has moved $\geq$ `GridDistancePoints` from last level

Emergency Close Triggered Unexpectedly

Review logs for trigger reason:

- Basket DD $\geq$ `EmergencyCloseAtDDPercent`
- Equity DD $\geq$ `MaxEquityDrawdownPercent`
- Free margin < 50% of `MinFreeMarginPercent`

Adjust parameters if triggers are too sensitive.

Important Notes

What This EA Does NOT Do

- ✗ Trade symbols other than XAUUSD
- ✗ Support MT5 (MT4 only)
- ✗ Use aggressive martingale (doubling lots)
- ✗ Set per-trade TP/SL on grid legs
- ✗ Manage multiple baskets simultaneously
- ✗ Mix BUY and SELL in one basket

Risk Disclaimer

Grid/recovery systems can experience significant drawdowns during strong trending markets. While this EA includes comprehensive risk controls, **no EA can guarantee profits or prevent all losses.**

- Always test thoroughly on demo first
- Never risk more than you can afford to lose
- Monitor the EA regularly, especially during news events
- Understand that past performance does not guarantee future results

Support & Modifications

This EA is delivered as-is with full source code. You may modify the code as needed, but ensure you understand the implications of any changes to risk management logic.

For questions about the implementation or to request modifications, refer back to the original requirements document (`requirements.txt`).

Version History

v1.0 (Initial Release)

- Complete grid/recovery implementation
- Three entry modes (MA, RSI, Time-based)
- Basket-level management
- Multi-layer risk controls
- Dynamic grid expansion

- Emergency close with slippage control
 - Volatility and spread filters
 - Trading hours filter
 - Optional trailing basket TP
 - Full logging and on-chart display
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Developed according to Netec1400 specifications

Priority: Survivability > Control > Stability > Profit